AQUEN FLUID TECHNOLOGIES LLP

Complete Water & Wastewater Treatment Solutions

Design | Manufacturing | Supply | Installation | Commissioning | AMC

EDI (ELECTRODEIONIZATION) UNIT

COMPLETE STAINLESS STEEL (SS-304 / SS-316) CONSTRUCTION

HIGH PURITY WATER FOR INDUSTRIAL APPLICATIONS

250 LPH TO 5,000 LPH

- 100% SS-304 / SS-316 construction
- Compact & skid mounted design
- Reliable & continuous operation
- Low operating cost
- Consistent high purity water
- Online monitoring & automation
- Custom designed as per requirement
- Suitable for RO, DM and ultrapure applications

EDI UNIT – 250 LPH



- Compact design
- Complete SS-304/316 construction
- Suitable for laboratories / small industries

EDI UNIT – 1,000 LPH



- Fully automatic with PLC
- High purity water output
- Compact skid mounted design

EDI UNIT – 2,500 LPH



- Modular design for high reliability
- Online monitoring (Conductivity, Flow, Pressure)
- Easy operation and maintenance

EDI UNIT – 5,000 LPH



- Heavy duty industrial design
- Continuous high purity water production
- Customized design as per site requirement

PROCESS FLOW DIAGRAM (TYPICAL – RO + EDI)



TYPICAL INLET & OUTLET PARAMETERS

Parameter	Unit	Typical Inlet (RO Permeate)	Typical Outlet (EDI Product)
pH	-	6.0 – 7.0	6.0 – 7.0
Conductivity	µS/cm	5 – 30	0.05 – 0.2
TDS	mg/L	3 – 20	< 0.1
Silica (SiO_2)	mg/L	< 1.0	< 0.01
Total Hardness (as $CaCO_3$)	mg/L	< 1.0	< 0.01
Chloride (Cl^-)	mg/L	< 1.0	< 0.01
Sulfate (SO_4^{2-})	mg/L	< 1.0	< 0.01
CO_2 (Carbon Dioxide)	mg/L	< 1.0	< 0.01
Iron (Fe)	mg/L	< 0.01	< 0.001
Manganese (Mn)	mg/L	< 0.01	< 0.001
Organics (TOC)	mg/L	< 0.5	< 0.1

KEY FEATURES & BENEFITS

- Complete SS-304 / SS-316 (Vessels, Piping, Fittings, Skid)
- EDI modules (Leading brands)
- Cartridge filters, high pressure pump, instruments in SS
- Produces high purity water (0.05 – 0.2 µS/cm)
- Continuous and chemical free operation (no acid/alkali regeneration)
- Compact and modular skid mounted design
- Automatic operation with PLC & HMI
- Online monitoring (Conductivity, Flow, Pressure)
- Low maintenance and long service life
- Pre-engineered and factory tested
- Customized design for 250 LPH to 5,000 LPH

APPLICATIONS


Pharmaceutical


Power Plant


Electronics / Semiconductor


Food & Beverage


Chemical Industry


Laboratory / Research

TECHNICAL SPECIFICATIONS (250 LPH TO 5,000 LPH)

Model	Capacity (LPH)	No. of EDI Modules	Feed Pump (SS) (LPH @ m)	Cartridge Filter (Size)	Inlet / Outlet Pipe Size (NB)	Skid Dimensions (L x W x H) mm (Approx.)	Connected Load (kW)
AQU-EDI-0.25	250	1	400 @ 30	5 Micron SS	15	1500 x 800 x 1500	1.0
AQU-EDI-0.5	500	1	800 @ 30	5 Micron SS	20	1800 x 900 x 1600	1.5
AQU-EDI-1.0	1,000	2	1,500 @ 35	5 Micron SS	25	2200 x 1000 x 1700	2.5
AQU-EDI-2.0	2,000	3	3,000 @ 35	5 Micron SS	32	2800 x 1200 x 2000	4.0
AQU-EDI-2.5	2,500	4	3,500 @ 40	5 Micron SS	40	3200 x 1200 x 2100	5.0
AQU-EDI-3.0	3,000	4	4,500 @ 40	5 Micron SS	50	3600 x 1500 x 2200	6.0
AQU-EDI-4.0	4,000	6	6,000 @ 40	5 Micron SS	50	4500 x 1800 x 2300	8.0
AQU-EDI-5.0	5,000	8	7,500 @ 40	5 Micron SS	65	5200 x 2000 x 2500	10.0

Note: Specifications are indicative and can be customized as per raw water quality, site conditions and brand of EDI modules.

MAJOR COMPONENTS (BOM)


EDI Module


SS Cartridge Filter


High Pressure Pump (SS)


SS Piping & Fittings


Instrumentation (Pressure Gauge, Flow Meter, Conductivity Meter)


PLC Panel with HMI (SS Enclosure)

STANDARD SCOPE OF SUPPLY

- EDI modules (as per capacity)
- Cartridge filter housing with filter elements
- High pressure pump (SS-304/316)
- All piping, fittings, valves (SS-304/316)
- Instrumentation (PG, Flow, Conductivity)
- SS skid with support
- Control panel with PLC & HMI
- Interconnecting cables & accessories
- Testing, commissioning & documentation

AQUEN FLUID TECHNOLOGIES LLP

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WTP | STP | ETP | ZLD | FABRICATION | SPARES | AMC | RETROFIT | AUTOMATION

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CLEANER WATER GREENER TOMORROW

AQUEN FLUID TECHNOLOGIES LLP

1. COMPLETE RO + EDI WATER TREATMENT SYSTEM

(Sand Filter + Carbon Filter + RO + EDI)



Raw Water (Inlet) → Sand Filter → Carbon Filter → Micron Filter → RO System → EDI Unit → Ultra Pure Water (Outlet)

TYPICAL INLET & OUTLET PARAMETERS

Parameter	Raw Water (Inlet)	RO Permeate (To EDI)	Product Water (Outlet)
pH	6.5 – 8.5	6.0 – 7.0	6.0 – 7.0
TDS	500 – 3,000 mg/L	< 50 – 200 mg/L	< 0.1 – 1.0 µS/cm
Conductivity	750 – 4,500 µS/cm	10 – 50 µS/cm	0.05 – 0.2 µS/cm
Silica (SiO_2)	< 20 mg/L	< 1 mg/L	< 0.01 mg/L
Hardness (as $CaCO_3$)	50 – 500 mg/L	< 1 mg/L	< 0.01 mg/L

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2. RO + EDI SYSTEM – 1,000 LPH

(SS-304 / SS-316)



TECHNICAL SPECIFICATIONS (1,000 LPH)

Parameter	Value
Flow Rate (Product Water)	1,000 LPH (1.0 m^3/hr)
RO Recovery	65 – 75 %
No. of RO Membranes (4040)	4 – 6 Nos.
EDI Modules	2 Nos. (as per design)
Installed Power	6 – 8 kW (approx.)
Skid Material	SS-304 / SS-316
Piping, Fittings, Valves	SS-304 / SS-316
Control Panel	SS Enclosure with PLC & HMI
Dimension (Approx.)	3000 (L) x 1200 (W) x 1800 (H) mm

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3. RO + EDI SYSTEM – 5,000 LPH

(SS-304 / SS-316)



TECHNICAL SPECIFICATIONS (5,000 LPH)

Parameter	Value
Flow Rate (Product Water)	5,000 LPH (5.0 m^3/hr)
RO Recovery	65 – 75 %
No. of RO Membranes (8040)	8 – 16 Nos.
EDI Modules	4 – 8 Nos. (as per design)
Installed Power	15 – 25 kW (approx.)
Skid Material	SS-304 / SS-316
Piping, Fittings, Valves	SS-304 / SS-316
Control Panel	SS Enclosure with PLC & HMI
Dimension (Approx.)	5500 (L) x 1800 (W) x 2200 (H) mm

4. PRETREATMENT SECTION (SAND FILTER + CARBON FILTER)



SPECIFICATIONS (PRETREATMENT FILTERS)

Parameter	Sand Filter (SS-304/316)	Carbon Filter (SS-304/316)
Flow Range	250 – 10,000 LPH	250 – 10,000 LPH
Vessel Type	Vertical Pressure Vessel	Vertical Pressure Vessel
Material	SS-304 / SS-316 (All Parts)	SS-304 / SS-316 (All Parts)
Media	Silica Sand	Activated Carbon (800 Iodine)
Operating Pressure	2 – 4 bar	2 – 4 bar
Inlet / Outlet Size (NB)	15 – 100 NB (as per capacity)	15 – 100 NB (as per capacity)

5. REVERSE OSMOSIS (RO) SYSTEM

(SS-304 / SS-316 – COMPLETE SS CONSTRUCTION)



RO SYSTEM SPECIFICATIONS (250 – 15,000 LPH)

Parameter	Value
Capacity Range	250 LPH to 15,000 LPH
Membrane Type	4040 / 8040 (as per capacity)
Membrane Housing	SS-304 / SS-316 (No FRP)
High Pressure Pump	SS Multistage Pump
Piping, Fittings, Valves	SS-304 / SS-316
Instrumentation	Pressure Gauges, Flow Meters, Conductivity, etc.
Control Panel	SS Enclosure with PLC & HMI
Skid	SS-304 / SS-316 – Compact Design

6. SINGLE EDI UNIT

(SS-304 / SS-316 – 250 LPH TO 5,000 LPH)



EDI UNIT SPECIFICATIONS

Parameter	Value
Flow Range	250 LPH to 5,000 LPH
EDI Module	1 – 8 Nos. (as per capacity)
Material (All Parts)	SS-304 / SS-316
Piping, Fittings, Valves	SS-304 / SS-316
Instrumentation	Pressure Gauge, Flow Meter, Conductivity Meter, etc.
Control Panel	SS Enclosure with PLC & HMI
Skid	SS-304 / SS-316 – Compact & Skid Mounted
Product Water Quality	0.05 – 0.2 µS/cm (Resistivity up to 15 – 18 MΩ-cm)